DESCRIPTION OF COURSEWORK

Course Code	EEE406
Course Name	Power Systems Analysis
Lecturer	Dr Abdul Rashid Abdul Rahman
Academic Session	09/2025
Assessment Title	Laboratory 3 - Transient stability studies

A. Introduction/ Situation/ Background Information

Objectives

To perform transient stability studies for the given tasks

Pre-Laboratory

Knowledge on fault analysis and familiarity with PowerWorld Simulator software is very important prior to conducting this laboratory. Therefore, students should complete the Pre-Laboratory steps below:

1. Study the power system stability lecture notes.
2. Install the PowerWorld Simulator software.
3. Learn on how to use PowerWorld Simulator software from "PowerWorld Simulator Quick Guide Ver 5".

Watch related tutorial video at powerworld.com or youtube.com websites for better understanding on using PowerWorld Simulator software.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Solve power system transient stability problems for continuous stable operation.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4) or five (5)** students's laboratory to be completed in 2 weeks. Simulation files(PowerWorld's Simulator pwb and pwd files) should be submitted with the report using filename"Lab3ReportGroupX" in softcopy (PDF) for every task that should follow the following format:

1. Introduction
2. Methodology
3. Results and Discussion
4. Conclusion
5. References

Due Date: **26th December 2024.**

Software Tool: PowerWorld Simulator software.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Create contingency condition of loss of generation that cause unstable condition in the electrical utility network. Analyze the generators MW, MVAR, speed and rotor angle before and after this contingency condition.	20
2.	Propose load shedding setting (load and maximum time) to mitigate the unstable condition in part (1). Analyze the generators MW, MVAR, speed and rotor angle after implementation of load shedding setting (load and maximum time).	30
3.	Create contingency condition of three phase transient fault on transmission line that cause unstable condition in the electrical utility network. Analyze the generators MW, MVAR, speed and rotor angle before and after this contingency condition.	20
4.	Propose critical clearing time to mitigate the unstable condition in part (3). Analyze the generators MW, MVAR, speed and rotor angle after implementation of critical clearing time.	30
	TOTAL	**100**

F. Task(s)

The task is to ensure that the electrical utility network operates with stable condition after major disturbances such as loss of generation, faults, line-switching operations, and sudden load changes. Following a disturbance, synchronous machine frequencies undergo transient deviations from synchronous frequency and machine power angles change. The objective of a transient stability study is to determine whether or not the machines will return to synchronous frequency with new steady-state power angles. Changes in power flows and bus voltages are also of concern. Table 1 provides the simulation parameters to be adopted.

Table 1

Slack bus	Bus 1
Simulation start time	0 second
Disturbance commence time	1 second
Simulation end time	10 seconds

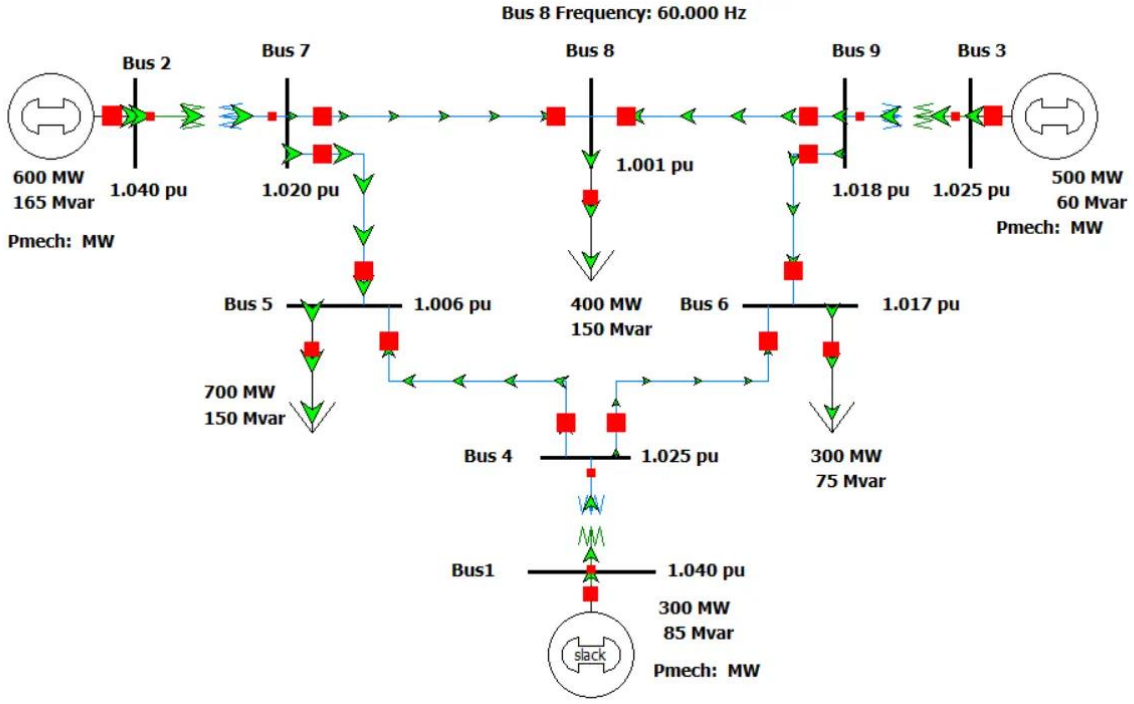


Figure 1 Electrical utility network diagram

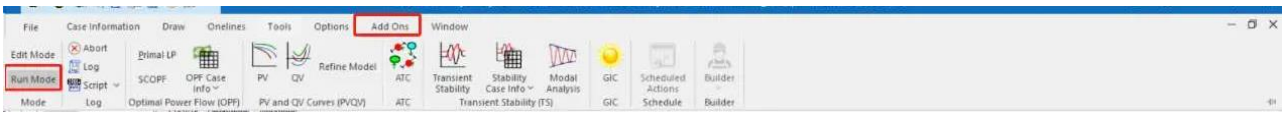
The power system depicted in the diagram is to be used for the transient stability analysis in this study. The entire network operates at a frequency of 60 Hz, a standard frequency for power systems in many regions.

Lab procedures/tasks

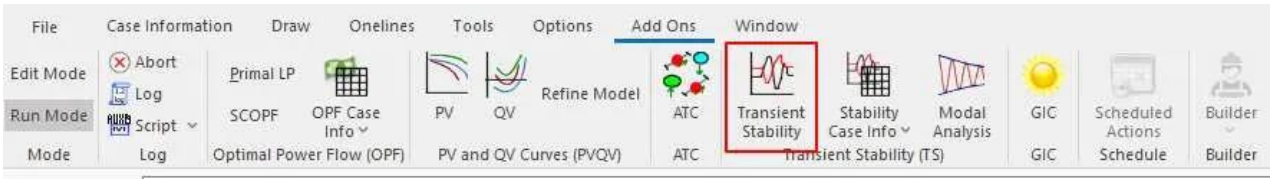
1. Create a contingency condition of loss of generation that cause unstable condition in the electrical utility network. Analyze the generators MW, MVAR, speed and rotor angle before and after this contingency condition.

<u>Contingency condition of the loss of generation</u>

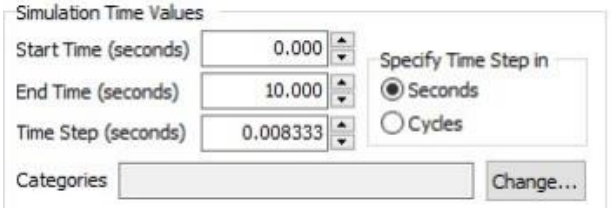
Step 1: Click “Run Mode” and “Add Ons”.



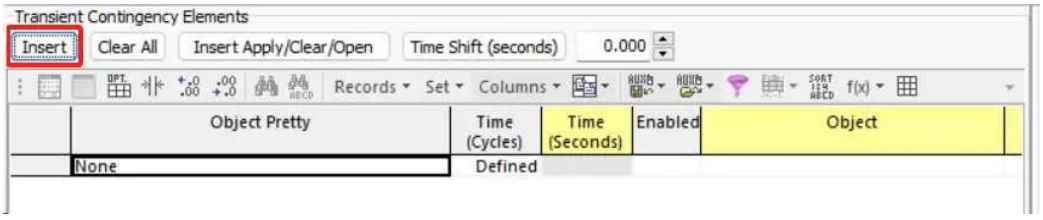
Step 2: Click “Transient Stability” and turn to the “Transient Stability Analysis”.



Step 3: At the simulation window, the “Start Time” to 0 and the “End Time” to be set to 10 seconds following the simulation parameters. The “Specify Time Step” should be changed in Seconds.



Step 4: Click , “Insert” at the Transient Contingency Elements window.



Step 5: Choose the Objective Type as “Generator”. Select the fault (e.g. at generator 2). Set the time to 1 seconds as the disturbance commence time. Set fault type as “Apply Fault” and “Open”.



2. Propose load shedding setting (load and maximum time (to the closest 0.01 second)) to mitigate the unstable condition in part (1). Analyze the generators MW, MVAR, speed and rotor angle after implementation of load shedding setting (load and maximum time).

Propose load shedding settings on 1.

Step 1: Based on Transient Contingency Elements setting on 1. Click 'Insert' button and a 20% load shedding should be added on load bus 5, an 80% load shedding should added on load bus 6. Set time 1.20s. (example)

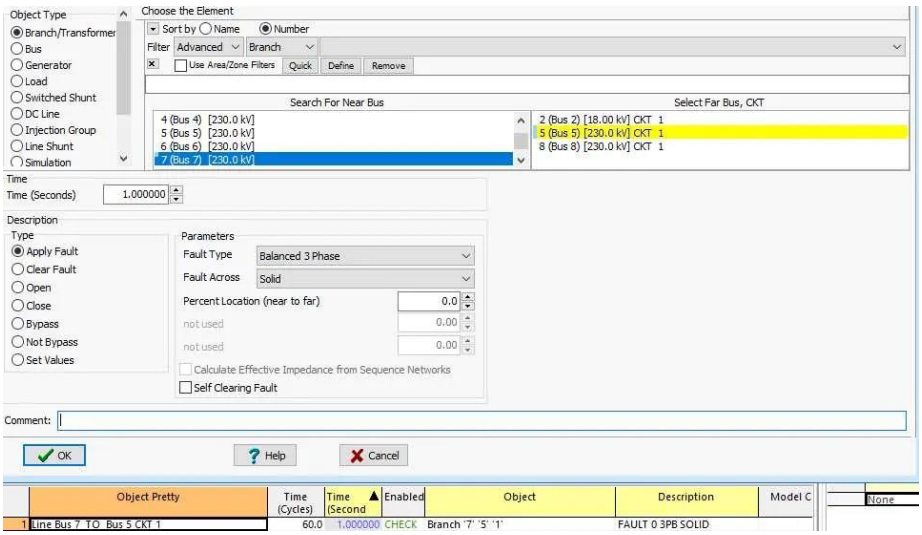
	Object Pretty	Time (Cycles)	Time (Second	Enabled	Object	Description	Model Criteria
1	Gen Bus 2 #1	60.0	1.000000	CHECK	Gen '2' '1'	OPEN	
2	Load Bus 5 #1	72.0	1.200000	CHECK	Load '5' '1'	SET Load 20 %	
3	Load Bus 6 #1	72.0	1.200000	CHECK	Load '6' '1'	SET Load 80 %	

3. Create a contingency condition of three phase transient fault on transmission line that cause unstable condition in the electrical utility network. Analyze the generators MW, MVAR, speed and rotor angle before and after this contingency condition.

Three phase transient fault on transmission line

Step 1 to Step 4: Follow steps in section 1

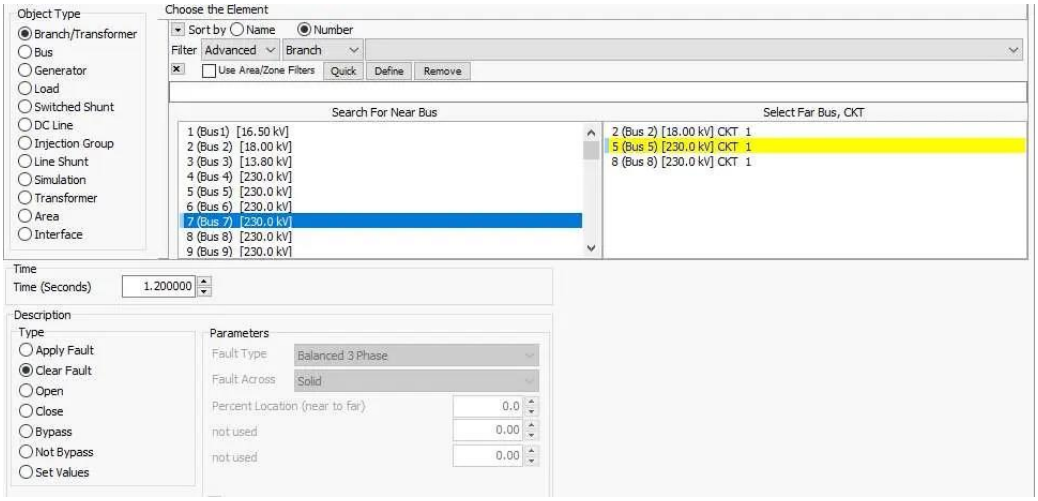
Step 5: Choose Objective type as "Branch/Transformer". Select the fault at transmission line between bus 7 to bus 5 (e.g.). Set the disturbance commence time to 1 seconds. Set Fault Type as "Apply Fault" and "Balanced 3 Phase", "Solid" as Fault Across, 0 as Percentage Location.



4. Propose critical clearing time (to the closest 0.01 second) to mitigate the unstable condition in part (3). Analyze the generators MW, MVAR, speed and rotor angle after implementation of critical clearing time.

Propose critical clearing time on 3.

Step 1: Based on Transient Contingency Elements setting on 3. Click 'Insert' button to add a clear fault on transmission line between bus 7 to bus 5. And set 1.20s.



5. Display the simulation snapshot results in your report.

APPENDIX 1

MARKING RUBRICS

Component Title	**Transient stability studies**				**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Contingency condition of loss of generation	**100% detail contingency created and simulated with comprehensive and correctly analyzed**	**80% detail contingency created and simulated with comprehensive and correctly analyzed**	**60% detail contingency created and simulated with comprehensive and correctly analyzed**	**40% detail contingency created and simulated with comprehensive and correctly analyzed**	**Less than 40% detail contingency created and simulated with comprehensive and correctly analyzed**	**4**	
Load shedding setting (load and maximum time) proposal to mitigate the loss of generation contingency	**100% detail load shedding setting (load and maximum time) proposed and simulated with comprehensive and correctly analyzed**	**80% detail load shedding setting (load and maximum time) proposed and simulated with comprehensive and correctly analyzed**	**60% detail load shedding setting (load and maximum time) proposed and simulated with comprehensive and correctly analyzed**	**40% detail load shedding setting (load and maximum time) proposed and simulated with comprehensive and correctly analyzed**	**less than 40% detail load shedding setting (load and maximum time) proposed and simulated with comprehensive and correctly analyzed**	**6**	
Contingency condition of three phase transient fault on transmission line	**100% detail contingency created and simulated with comprehensive and correctly analyzed**	**80% detail contingency created and simulated with comprehensive and correctly analyzed**	**60% detail contingency created and simulated with comprehensive and correctly analyzed**	**40% detail contingency created and simulated with comprehensive and correctly analyzed**	**less than 40% detail contingency created and simulated with comprehensive and correctly analyzed**	**4**	
Critical clearing time proposal to mitigate the three phase transient fault on transmission line contingency	**100% detail critical clearing time proposed and simulated with comprehensive and correctly analyzed.**	**80% detail critical clearing time proposed and simulated with comprehensive and correctly analyzed.**	**60% detail critical clearing time proposed and simulated with comprehensive and correctly analyzed.**	**40% detail critical clearing time proposed and simulated with comprehensive and correctly analyzed.**	**less than 40% detail critical clearing time proposed and simulated with comprehensive and correctly analyzed.**	**6**	
					TOTAL		

Note to students: Please include the marking rubric when submitting your coursework.